Data Request Tables

The proposed Data Science project will predict whether an order will be returned using information available at or before the point of purchase. The requested data should therefore include sufficient historical information to construct predictive features while also providing the return outcome required to train and evaluate the classification model.

All predictive variables must represent information available at the prediction point. Variables generated after purchase, delivery or return initiation must not be used as model predictors. Such variables may still be requested where they are required for target construction, model evaluation or post-hoc analysis. Historical behavioural, seller and operational measures should be calculated using information available before the prediction point to prevent data leakage.

Target Variable

The target variable for the eventual Data Science project will be:

Target variable	Class	Encoded value	Meaning
Return Status	Returned	1	The order was returned.
Return Status	Not Returned	0	The order was not returned.

The target variable is Return Status, which creates a binary classification problem with two possible outcomes: Returned (1) and Not Returned (0). The unit of prediction is the order. An order will be classified as Returned (1) if it meets the agreed definition of a returned order within the specified return observation window. Otherwise, it will be classified as Not Returned (0).

For orders containing multiple items, STADIOalot should confirm the appropriate order-level aggregation rule. For example, an order may be classified as Returned (1) if at least one item from the order is returned. The final definition and observation window should be agreed with STADIOalot before model development.

Table 1: Orders and Transactions

Column name	Data type	Example	Additional information
Order ID	String	ORD849302	Unique identifier for each order.
Customer ID	String	CUST29482	Pseudonymised customer identifier that can be linked across datasets
Order date/time	Datetime	2026-04-15 14:32:10	Use consistent yyyy-MM-dd HH:mm:ss format
SKU	String	SKU183920	Unique product identifier. If an order contains multiple SKUs, provide order-item records linked using Order ID.
Seller ID	String	SELL4839	Marketplace seller identifier; identify first-party orders appropriately
Quantity	Integer	2	Number of units purchased for the specific order-item. Values should be greater than or equal to 1.
Unit price	Numeric (Decimal)	899.00	Price per unit at the time the order was placed.
Promotion amount	Numeric (Decimal)	100.00	Discount or promotional amount applied to the order at the time of purchase.
Payment method	Categorical (String)	Card	Use a controlled list of payment methods
Delivery option	Categorical (String)	Standard	Delivery method selected by the customer when placing the order. Use a consistent controlled list.
Customer province	Categorical (String)	GP	Province associated with the customer's delivery or registered address. Use standard South African province codes.

Table 2: Returns

The Returns dataset is required primarily to construct the target variable, evaluate model performance and analyse the characteristics and costs of returned orders. Post-purchase return information must not be used as input features for the predictive model.

Column name	Data type	Example	Additional information
Return ID	String	RET55081	Unique identifier for each return event.
Order ID	String	ORD849302	Links the return to the original order.
Return date/time	Datetime	2026-04-20 09:15:00	Date/time the return was initiated. If available, also provide return receipt date/time separately. Post-purchase; not a predictive feature.
Returned quantity	Integer	1	Number of units returned. Used for outcome analysis/target construction; not a predictive feature.
Return reason	Categorical (String)	Wrong size	Standardised return reason. Known only after return and must not be used as a predictive feature.
Return condition	Categorical (String)	Unopened	Controlled list of return conditions. Post-return information; not a predictive feature.
Refund amount	Numeric (Decimal)	899.00	Refund amount in ZAR. Post-purchase information; not a predictive feature.
Refund outcome	Categorical (String)	Approved	Controlled refund outcome. Post-purchase information; not a predictive feature.
Return cost	Numeric (Decimal)	65.00	Processing/reverse-logistics cost in ZAR. Useful for business impact analysis; not a predictive feature.
Avoidable return flag	Binary	1	Existing business-defined classification if available. If not available, definition should be agreed with STADIOalot. Not a predictive feature unless known at prediction time.

Table 3: Product Catalogue

Product ratings and review counts must reflect information available before the order was placed, not information added afterwards.

Column name	Data type	Example	Additional information
SKU (Product ID)	String	SKU183920	Unique product identifier used to link product catalogue information to orders and inventory records.
Product category	Categorical, String	Apparel	Use standard category hierarchy
Subcategory	Categorical (String)	Women's shoes	Standardised subcategory
Brand	Categorical (String)	Brand A	Consistent brand identifier
Size	Categorical (String)	Large	Required where applicable
Colour	Categorical (String)	Black	Standardised colour
Product price	Numeric (Decimal)	899.00	Historical product price at the time of purchase.
Product rating	Numeric (Decimal)	4.3	Rating available at the prediction point. Document the rating scale, e.g. 0–5.
Review count	Integer	128	Number of reviews available before the prediction point.
Listing age days	Integer	84	Number of days the product had been listed at the prediction point. Must be ≥ 0.

Table 4: Marketplace Sellers

Column name	Data type	Example	Additional information
Seller ID	String	SELL4839	Unique seller identifier
Seller rating	Numeric (Decimal)	4.2	Seller rating available at the prediction point. Document the rating scale.
Seller tenure months	Integer	18	Time seller has been active
Average shipping days	Numeric, Decimal	2.8	Historical average shipping time calculated using orders before the prediction point.
Late shipment rate	Numeric, Decimal	0.07	Historical proportion of previous shipments classified as late, calculated before the prediction point.
Complaint count	Integer	12	Number of seller complaints during a defined historical period before the prediction point.

Table 5: Customer Information

Column name	Data type	Example	Additional information
Customer ID	String	CUST29482	Pseudonymised unique customer identifier; must link to the Orders and Customer Behaviour datasets
Customer age	Integer	25	Age of customer at the time of the order
Customer gender	Categorical (String)	Female	Use controlled values; include an Unknown/Not disclosed and Other category where applicable
Customer province	Categorical (String)	GP	South African province code associated with the customer's location at the prediction point.
Account creation date	Datetime	2023-06-14 10:25:00	Date and time the customer account was created
Customer tenure months	Integer	36	Customer tenure calculated as at the prediction point.
Preferred shopping channel	Categorical (String)	Mobile App	Channels such as Website, Mobile App or Other
Average order value	Numeric, (Decimal)	1,245.50	Historical average order value calculated using eligible orders before the prediction point, in ZAR.
Total historical spend	Numeric, (Decimal)	38,615.50	Cumulative customer spend before the prediction point, in ZAR.
Previous returned orders	Integer	2	Number of orders returned by the customer before the prediction point.

Table 6: Customer Behaviour

Column name	Data type	Example	Additional information
Customer ID	String	CUST29482	Pseudonymised identifier linking behaviour to the customer.
Look-back period	Integer	30	Number of days over which behavioural activity is measured. The period should end at or before the prediction point.
Sessions	Integer	24	Number of shopping sessions during the defined look-back period before purchase.
Product views	Integer	183	Number of product detail views during the look-back period before purchase.
Add-to-basket count	Integer	19	Number of add-to-basket events during the look-back period before purchase.
Previous orders	Integer	31	Number of completed orders before the prediction point.
Previous returns	Integer	6	Number of previous returned orders before the prediction point.
Customer return rate	Numeric, (Decimal)	0.19	Previous returns divided by eligible previous orders. Calculated using information available before the prediction point.

Table 7: Inventory and Warehouse

Column name	Data type	Example	Additional information
SKU	String	SKU183920	Product identifier linked to the order/product catalogue.
Distribution centre	Categorical (String)	DC3	Distribution centre associated with fulfilment. Where possible, provide the centre known/assigned at the prediction point.
Stock on hand	Integer	240	Units available at the prediction point. Must be ≥ 0.
Stock age days	Integer	46	Age of available stock at the prediction point. Must be ≥ 0.
Historical pick/pack time	Numeric, (Decimal)	7.5	Historical average pick/pack time, preferably calculated from previous orders before the prediction point. Unit: minutes.
Inbound date	Datetime	2026-03-01 08:00:00	Date/time the relevant stock was received into the distribution centre. Must be available at or before the prediction point.

Table 8: Delivery and Logistics

Column name	Data type	Example	Additional information
Order ID	String	ORD849302	Links logistics information to the order.
Estimated delivery date/time	Datetime	2026-04-18 16:20:00	Estimated delivery information available at the time of purchase. Can be used as a predictive feature if available.
Delivery option	Categorical (String)	Standard	Delivery option selected at purchase.
Delivery province	Categorical (String)	KZN	Destination province associated with the order.
Route distance km	Numeric, (Decimal)	18.4	Estimated/recorded route distance available at the prediction point, where applicable.
Historical delivery time	Numeric, (Decimal)	2.4	Historical average delivery duration for the relevant route/region, calculated before the prediction point. Unit: days.
Historical delivery failure rate	Numeric, (Decimal)	0.06	Historical proportion of failed/late deliveries calculated before the prediction point.
Actual delivery date/time	Datetime	2026-04-19 11:45:00	Actual delivery timestamp. Post-purchase; use for evaluation/analysis only, not prediction.
Delivery attempts	Integer	2	Number of attempts for the current order. Post-purchase; not a predictive feature.
On-time flag	Binary	0	1 = on time, 0 = late. Post-purchase; not a predictive feature.